Graphical Modeling

→ Graphical Modeling is a mathematical framework used to represent complex relation-
-ship between variables using graph.

→ widely used in statistics, machine learning, AI and decision making processes.

A graphical model is a way to represent complex probability distributions using graphs.
It helps in understanding and computing relationship between multiple variables efficiently.

A GM is a probabilistic model where a graph represents dependencies among variables.

node → variable

edge → dependency b/w variables.

eg - medical diagnosis

Node → disease and symptoms.

Edges → show which symptoms are related to which disease.

Instead of using complex joint probability distribu-tion over all variables, graphical model allow us to simplify calculations by representing only direct dependencies.

Types of Graphical Models :

2 main types : 1) Bayesian Networks (Directed Graphical Models)

2) Markov Random fields (MRFs) (Undirected Graphical Model)

Bayesian Networks (Direct Graphical Models)

Bayesian Network (BNs) use directed edges (arrows) to represent casual (or) conditional dependencies b/w variables.

Characteristics :

- uses Directed Acyclic Graph (DAG
- each node represents a random variable.
- Each directed edge represent a conditional dependency.

eg → consider a Bayesian Network for medical diagnosis :

flu → fever → Test Result

flu → Cough

having flu ↑↑ prob. of having cough and fever.

the result depend on fever.

The prob of fever depends of flu :

$P(\text{fever} / \text{flu})$

Bayesian n/w uses Baye's theorem for probability inference.

Advantages:

- Captures casual relationships.
- Support efficient inference using conditional independence.
- useful for decision making [eg - fraud detection, medical diagnosis]

Markov Random fields (MRFs) (Undirected - Graphical Models)

MRF uses undirected edges, meaning they do not assume direct casuality, only mutual dependence.

Characteristics:

- uses undirected graphs
- each node is a random variable
- edge show statistical dependencies not casuality.
- uses clique (fully connected subgraphs) to define relationships.

eg - A markov Random field for image processing

pixel_1 - Pixel_2 - Pixel_3
| | |
pixel_4 - Pixel_5 - Pixel_6

→ each pixel's brightness is influenced by neighboring pixels | → No casual direction - just mutual dependence

MRF use the Markov Property, meaning a node is independent of all other nodes except its neighbor

Advantages:

- useful for problems where dependencies are symmetric (eg - image segmentation, social n/w)
- Supports local computations.

Features	Bayesian N/W	Markov Random fields
Graph type	Directed (DAG)	Undirected
Dependencies	Casual relationship	Mutual dependence
Independence	conditional Independence	Markov Property
Inference	Bayes Rule	factor graph
Applications	Decision making NLP, Medical Diag.	Image processing social n/w

Applications of graphical model:

1. ML and AI
2. Computer vision
3. Medical Diagnosis
4. finance & Business

Key Concepts in GM

a) Conditional Independence :

CI means that two variables are independent given a third variable.

eg - suppose we have three variable Rain, wet Grass and Sprinkler.

If we know the Rain status, then wet grass doesnot depend on the sprinkler.

$$P\left(\frac{\text{Wet Grass}}{\text{Rain, Sprinkler}}\right) = P\left(\frac{\text{Wet Grass}}{\text{Rain}}\right)$$

this help in simplifying probability calculations.

b) Inference in Graphical Models :

Inference means computing the probability of unknown variables given known values.

Inference method — 1. Exact Inference
2. Approximate Inference

Exact Inference	Approximate Inference
• Variable Elimination : Removing irrelevant variables step-by-step.	• Monte Carlo Sampling : estimating prob. using repeated random sampling.
• Junction Tree Algorithm : Converting the graph into a tree to perform efficient computations.	• Loopy Belief Propagation : An iterative method of graph with cycles

Learning in Graphical Models:

Learning involves finding the structure and parameters of a graphical model from data.

1. **Structure Learning**

Discovering the graph (node & edges) from data.

algo - greedy - search
constraint - based method.

2. **Parameter learning**

Estimating probabilities from observed data.

Method : Maximum Likelihood Estimation (MLE), Expectation - Maximization Em

eg - if given a dataset of medical records, which symptoms are related to which disease the prob of getting a disease given certain symptoms.

Binary Decision Diagram

→ BDD is a data structure used to represent Boolean functions efficiently.

→ It is widely used in graphical models, probabilistic inference especially in areas like, model checking and optimization.

A BDD is a directed acyclic graph (DAG) that represents a Boolean function. It provides a compact and efficient way to encode Logical expressions.

eg — consider the boolean function

$$F(A, B, C) = (A \wedge B) \vee (\sim A \wedge C)$$

↓ Instead of storing this function in a truth table (which requireg 2^n entires for n variable), we use a BDD for a more effient representation.

[illegible]

2 types of BDD —

1 Ordered BDD

- variable must appear in the same order along every path.
- uniqueand compact Representation.

2. Reduced Ordered BDD

- An optimized version of OBDD.
- Redundant nodes and duplicate subgraph are removed.
- Ensures minimal representation

$F(A, B, C) = (A \wedge B) \vee (\neg A \wedge C)$

Steps to construct BDD:

1. <u>Identify Variables</u> : 3 variable - A, B, C

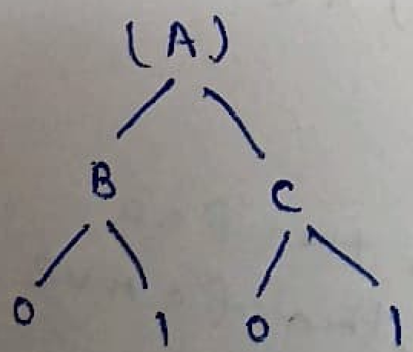
2. <u>Decision Order</u> : we chose the variable order
$A \to B \to C$

3. <u>Creates Nodes</u> : each variable A, B, C is a decision nodes.

4. <u>Determine Terminal Nodes</u> : The o/p are either 0 (false) or 1 (true)

5. <u>Draw Edges</u> :

- $A = 1 \to$ follow B's decision (since $A \wedge B$)
- $A = 0 \to$ follow C's decision (since $\neg A \wedge C$)



If A = 1, then B decides the output

$B = 1 \to$ o/p 1

$B = 0 \to$ o/p 0

If A = 0, then C decides the output

$C = 1 \to$ o/p = 1

$C = 0 \to$ o/p = 0

This is reduced order BDD (ROBDD) because redundant nodes are removed.

Q. Construct BDD for the function

$f(A, B) = A \oplus B$

(XOR function : o/p 1 when i/p differs)

1. Variable - A, B
2. Decision order - $A \rightarrow B$
3. Determine terminal values :

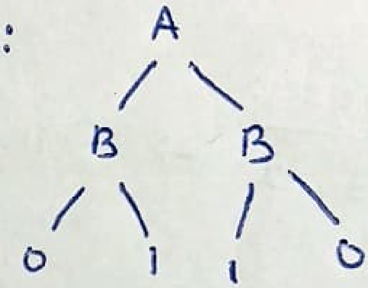
If $A = 0$, $B = 0 \rightarrow f = 0$

$A = 0$ $B = 1 \rightarrow f = 1$

$A = 1$ $B = 0 \rightarrow f = 1$

$A = 1$ $B = 1 \rightarrow f = 0$

BDD Representation:



When $A = 0$, then function depends only on B.
When $A = 1$, functn depens only on B.

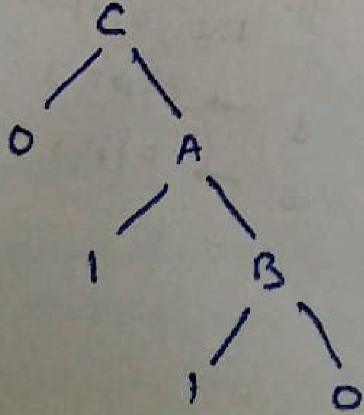
Q. Construct BDD for

$f(A, B, C) = (A \vee B) \wedge C$

1. Variables = A, B, C
2. Truth table representation -

If $C = 0$, then $f = 0$

If $C = 1$, then $f = A \vee B$



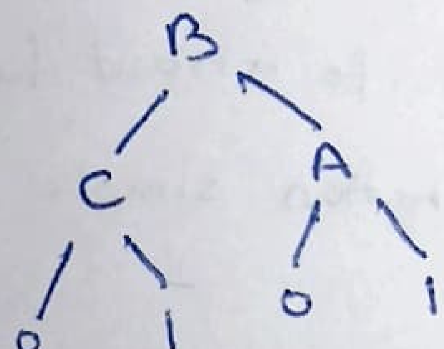
Q. Construct a BDD for the function $f(A, B, C) = (A \wedge B) \vee (B \wedge C)$

1. variables : A, B, C
2. Truth table analysis :

function true if either:

A and B are both 1

B and C are both 1



B = 0 leads 0 or C

B = 1 leads A

Q. What is the worst-case size of a BDD with n variables?

In worst case, BDT has $2^n - 1$ nodes.

Markov Property:

It states that:
The future state of a system depends only on its present state and not on the states (past)

In other words,
given the present state, the past does not provides any additional info about future.

This makes prediction and computation simple in probabilistic models.

Mathematical definition:

A stochastic process (X_t) satisfy the MP if

$$\boxed{P(X_{t+1} \mid X_t, X_{t-1}, \ldots X_1) = P(X_{t+1} \mid X_t)}$$

↳ only depend on present state.

example of MP — Random walk
Weather Prediction